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Research Communications

E75-induced Toll/NF- κ B signaling cooperates with Notch and Hippo pathways to promote tumor malignancy

Tumors are defined by uncontrolled cell proliferation (Hariharan and Bilder, 2006). Benign tumors are typically slow-growing and localized, while malignant ones are invasive and aggressive. The nuclear receptor *Eip75B* (*E75*), a heme-binding protein responsive to ecdysone signaling, encodes three major isoforms, *E75A*, *E75B*, and *E75C* (Bialecki et al., 2002), among them, only *E75A* and *E75C* contain zinc finger domains that enable DNA binding. *E75* regulates multiple key biological processes, such as cell growth, migration, metabolism, and differentiation (Reinking et al., 2005; Ahmed et al., 2020; Wang et al., 2020). Our prior work demonstrated that *E75* promotes tumor aggressiveness by suppressing the Hippo pathway and activating Notch signaling (Wang et al., 2024). However, we also found that even the combined Notch activation and Hippo inhibition by overexpressing the active form of Notch and Yki is insufficient to induce malignancy, suggesting the involvement of additional pathways. To further investigate the underlying mechanisms, we reanalyzed the bulk RNA-seq data of benign (*scrib*^{-/-}, *wts*^{-/-}) and malignant (*E75* + *scrib*^{-/-}, *wts*^{-/-}) tumors from our earlier work and identified significant enrichment and activation of the Toll signaling-related genes (Figs. 1A, 1B, S1A).

The Toll pathway, initially identified in embryonic dorsal-ventral patterning (Nussleinvolhard and Wieschaus, 1980) and innate immunity (Lemaitre et al., 1996), has recently been implicated in tumor growth in larvae (Brutscher et al., 2025; Dillard et al., 2025) while preventing proliferation of injected oncogenic cells in adult flies (Ji and Hoffmann, 2024), highlighting its role in tumorigenesis. In *Drosophila*, canonical Toll signaling is activated by Spätzle (Spz) binding to the Toll receptor, triggering a cascade via dMyD88, Tube, and Pelle, leading to nuclear translocation of NF- κ B factors Dorsal (dl)/Dorsal-related immunity factor (Dif). These induce immune effectors like Drosomycin (Drs), Attacin, and Cecropin (Cec) (Valanne et al., 2011). We found that ectopic *E75* expression in *scrib*^{-/-}, *wts*^{-/-} tumors upregulated key Toll components, including *spz*, *dorsal*, and *Drs-GFP* (Fig. S1B–S1D), indicating the activation of the Toll pathway. Next, we extended our analysis to physiological contexts to determine whether *E75* similarly regulates Toll signaling in normal development. Bulk RNA-seq analysis of *E75* overexpressed wing disc enriched Toll-related gene expression (Figs. 1C, 1D, S2A), upregulated *spz*, *dl*, *Dif-GFP*, and *Drs-GFP* (Figs. 1E–1G', S1E, S1F), and reduced the negative regulator *cactus* (Fig. S1G and S1H). qPCR confirmed elevated mRNA levels of *Peptidoglycan Recognition Protein* (*PGRP*), *Spz*, *Toll*, *dl*, *Dif*, *Cec*, and *Drs*, with reduced *cactus* (Fig. S1I). Conversely, *E75* knockdown

downregulated *spz*, *dl*, and *Drs-GFP*, while increasing *cactus* (Fig. S2B–S2E'), demonstrating a role for *E75* in the positive regulation of Toll signaling during the development and in cancer.

To investigate the underlying mechanism, we leveraged the DNA-binding ability of *E75* (Wang et al., 2024). Reanalysis of *E75* CUT&Tag data revealed binding at promoters of *spz*, *Tl*, *dl*, and *Dif* (Fig. 1H), but minimal occupancy at downstream effectors—unlike strong binding at the promoter region of *Myc* (Fig. S1J)—suggesting *E75* primarily activates upstream regulators of Toll pathway. Luciferase assays in *Drosophila* S2 cell confirmed that *E75* enhances transcription via these promoter regions (Fig. 1I and 1J). Hemin, an *E75* ligand, further boosted *E75*-mediated activation of *Dif* (Fig. 1K), identifying *E75* as a direct, ligand-enhanced activator of Toll pathway genes.

Overexpressing *dl* or *Dif* by mosaic analysis with a repressible cell marker (MARCM) clones exhibited significantly increased clonal size (Fig. S3A and S3B) and synergized with *scrib*^{-/-}, *wts*^{-/-} to promote tumor overgrowth (Fig. S3C and S3D). However, unlike *E75*-driven tumors, these did not block pupation (Fig. S3E), indicating that while Toll pathway activation enhances tumor growth, it is insufficient on its own to drive malignant progression. Furthermore, we found that knockdown of *dl* or *Dif* in *E75* + *scrib*^{-/-}, *wts*^{-/-} tumors rescued pupation defects and suppressed overgrowth (Figs. 1L–1N S3F–S3I), confirming that Toll signaling mediates *E75*-induced tumor malignancy. Given *Mmp1*'s role downstream of Toll (Zhao et al., 2025) and its upregulation in *E75* + *scrib*^{-/-}, *wts*^{-/-} tumors (Wang et al., 2024), we tested dependence on Toll signaling. *Dif* + *scrib*^{-/-}, *wts*^{-/-} tumors showed elevated *Mmp1* (Fig. S3J and S3K), while *Dif* knockdown reduced *Mmp1* (Fig. S3L and S3M), linking Toll to *E75*-driven *Mmp1* induction. As *E75* + *scrib*^{-/-}, *wts*^{-/-} tumors invade the ventral nerve cord (VNC) (Wang et al., 2024), and Toll promotes invasion in other models (Dillard et al., 2025), we tested whether Toll mediates *E75*-induced invasion. Indeed, *dl/Dif* knockdown reduced proliferation and invasion (Fig. 1O–1R), indicating that Toll pathway activation drives tumor growth and invasiveness, though not full malignancy alone.

To further characterize the role of *E75*, we employed a benign *N^{act}* + *Yki* (NY) tumor model. Co-expression of *E75* with NY (*E75* + *N^{act}* + *Yki*: ENY) induced malignant transformation, resulting in enlarged tumors and failed pupation (Fig. S4A and S4B). Paradoxically, despite *E75* reducing NICD/Yki binding at target gene transcription start sites (TSSs) (Wang et al., 2024), tumor growth

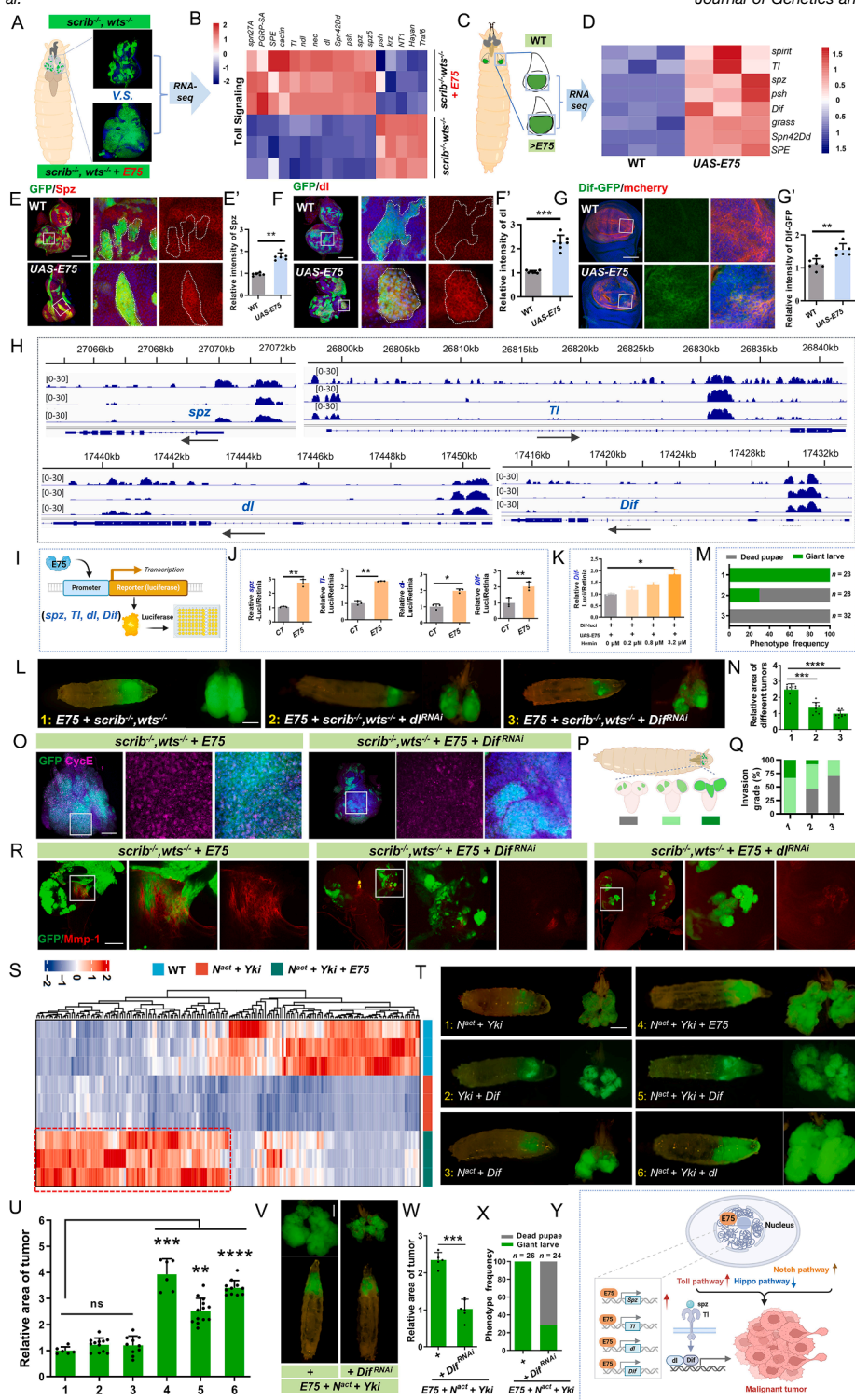


Fig. 1. E75-induced Toll activation cooperates with Notch and Hippo pathways to promote tumor malignancy. **A:** Schematic of *Drosophila* eye-antennal discs with *ey-Flp*-MARCM-induced mosaics. **B:** Heatmap of Toll pathway genes in *scrib*^{-/-}, *wts*^{-/-} and *E75* + *scrib*^{-/-}, *wts*^{-/-} tumors. **C:** Schematic representation of wing discs from larvae WT and *E75* overexpression induced by *nub*-Gal4. **D:** Heatmap of Toll signaling-related genes in WT and *E75* overexpression wing discs. **E–G:** Confocal images and quantification of Spz (**E** and **E'**), Dorsal (**F** and **F'**), and Dif-GFP (**G** and **G'**) levels. **H:** E75 CUT&Tag signals at Toll pathway gene promoters. Dark blue represents the peaks. **I:** Schematic of luciferase assay. **J:** Luciferase activity of *spz*, *dl*, *Dif*, and *Ti* promoters upon *E75* expression. **K:** Dif-Luc activity with increasing hemin. **L–N:** Larval images (**L**), pupation rates (**M**), and tumor area (**N**) in control and *E75* + *scrib*^{-/-}, *wts*^{-/-} tumors. **O–R:** Tumor proliferation (CycE, **O**), invasion grading (**P** and **Q**), and Mmp1 staining in VNC (**R**). **S:** Heatmap of gene expression in WT, NY (*N^{act}* + *Yki*), and ENY (*N^{act}* + *Yki* + *E75*) tumors. **T–X:** Confocal images of corresponding tumors (**T** and **V**) with quantification of relative tumor size (**U** and **W**), pupation ratio (**X**). **Y:** Schematic illustration depicting how *E75*-induced activation of the Toll pathway synergizes with the Hippo and Notch pathways to promote tumor malignancy. Data are presented as mean ± SD. Differences were determined by two-tailed Student's *t*-test (**E'**, **F'**, **G'**, **J**, **W**) and one-way ANOVA with mixed-effects (**K**, **N**, **U**). *, *P* < 0.05; **, *P* < 0.01; ***, *P* < 0.001; ****, *P* < 0.0001; ns, not significant, *P* > 0.05. Scale bars, 100 μm (**E**, **F**, **G**, **O**, **R**); 200 μm (**L**, **T**, **V**). WT, wild-type; VNC, ventral nerve cord; SD, standard deviation.

increased, suggesting that NICD/Yki may bind alternative genomic sites. CUT&Tag analysis revealed 1599 ENY-specific NICD/Yki peaks (Fig. S4C). RNA-seq analysis identified 1311 genes upregulated in ENY; overlapping with peaks yielded 175 candidate genes (Figs. 1S and S4C). Analysis of the associated 300 peaks using Hypergeometric Optimization of Motif Enrichment (HOMER) and Find Individual Motif Occurrences (FIMO) showed motif enrichment for Toll/Imd pathway transcription factors, including dl, Relish (Rel), and Dif (Fig. S4D), and a *de novo* E75 motif in 12.33%–25.0% of peaks (Fig. S4E). KEGG analysis confirmed Toll/Imd signaling as a top enriched pathway (Fig. S4F), implicating it in E75-driven tumor malignancy.

Next, to elucidate the cooperative interplay among the Notch, Hippo, and Toll pathways in malignant transformation, we conducted genetic interaction studies by co-expressing activated forms or transcriptional regulators of these signaling components, namely N^{act} , Yki, dl, and Dif, in larval eye discs. By day 6 after egg laying, tumors induced by Yki + Dif or N^{act} + Dif were comparable in size to those driven by N^{act} + Yki (Fig. 1T). However, Yki + Dif and N^{act} + Dif co-expression resulted in markedly higher pupation failure rates, 60% and 41.8%, respectively, compared with NY (Fig. S4G), indicating a more aggressive, malignant phenotype despite similar tumor burden. To further assess potential synergistic effects, we generated tumor models co-expressing N^{act} + Yki + Dif and N^{act} + Yki + dl. Strikingly, both triple combinations recapitulated the fully malignant phenotype observed in E75 + N^{act} + Yki tumors, characterized by substantial tumor enlargement and complete pupation failure (100%) (Figs. 1T, 1U, S4G). These findings demonstrate that activation of the Toll pathway synergizes with Notch and Hippo signaling to drive malignant progression. Finally, to establish a direct link between Toll signaling and ENY-mediated tumorigenesis, we knocked down Dif in ENY tumors. Dif knockdown significantly reduced tumor size and partially restored pupation rates (Fig. 1V–1X), confirming the essential role of Toll signaling in E75-driven malignant transformation.

In summary, we demonstrate that E75 directly binds and activates core Toll pathway components, driving tumor malignancy in *Drosophila* imaginal discs through an unrecognized synergistic interaction among Toll, Notch, and Hippo pathways, thereby promoting aggressive tumor phenotypes (Fig. 1Y). However, since imaginal discs are not canonical immune tissues in *Drosophila*, the physiological function of E75 in bona fide immune organs—such as the fat body and trachea—requests further investigation. Moreover, the mammalian homolog of E75, REV-ERB α , appears to regulate NF- κ B signaling in a manner distinct from E75-mediated Toll activation in *Drosophila*. Evidence suggests that REV-ERB α suppresses NF- κ B in microglia and human endometrial stromal cells (Griffin et al., 2019; Guo et al., 2019), whereas its loss in primary astrocytes does not affect basal NF- κ B activation (Griffin et al., 2019), indicating cell type-specific and context-dependent differences.

Data availability

The raw data of RNA sequencing in this study have been deposited in the National Center for Biotechnology Information Sequence Read Archive, with the accession number BioProject: PRJNA1067005. The other raw data can be requested from the corresponding author through a reasonable inquiry.

Conflict of interest

The authors declare no competing interests.

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Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jgg.2025.10.006>.

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